

AIDOS LAB
AI FOR DATA-ORIENTED SCIENCE

UNI FR

Geometric and topological methods in deep learning

XIV Encuentro de Jóvenes Topólogos — Logroño, Universidad de La Rioja

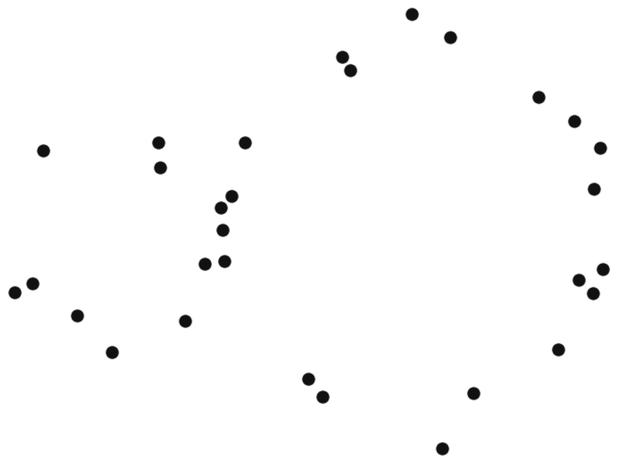
Inés García Redondo — 20/10/2026

Data has shape, shape has meaning, and meaning begets understanding

Gunnar Carlsson, paraphrased

Topological Data Analysis

Take inspiration from topology to define data analysis techniques and descriptors that capture *shape features* from data, are *amenable to statistical analysis* and are r*obust to noise* perturbations





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Take inspiration from topology to define data analysis techniques and descriptors that capture *shape features* from data, are *amenable to statistical analysis* and are r*obust to noise* perturbations

Topological Deep Learning

Learning on topological domains

How can networks process higher-order relational data?

Incorporating topology into learning

How can topological information guide a model?

Understanding networks through topology

What can shape reveal about learning and model behaviour?

Papamarkou, T., et al. (2024). "Position: Topological Deep Learning is the New Frontier for Relational Learning". *Proceedings of the 41st International Conference on Machine Learning*, 39529–39555.

Outline of the course

Session 1

Introduction to Deep Learning and Persistent Homology

Session 2

Infusing learning with topology and linking topology and generalization

Session 3

The topology of internal representations

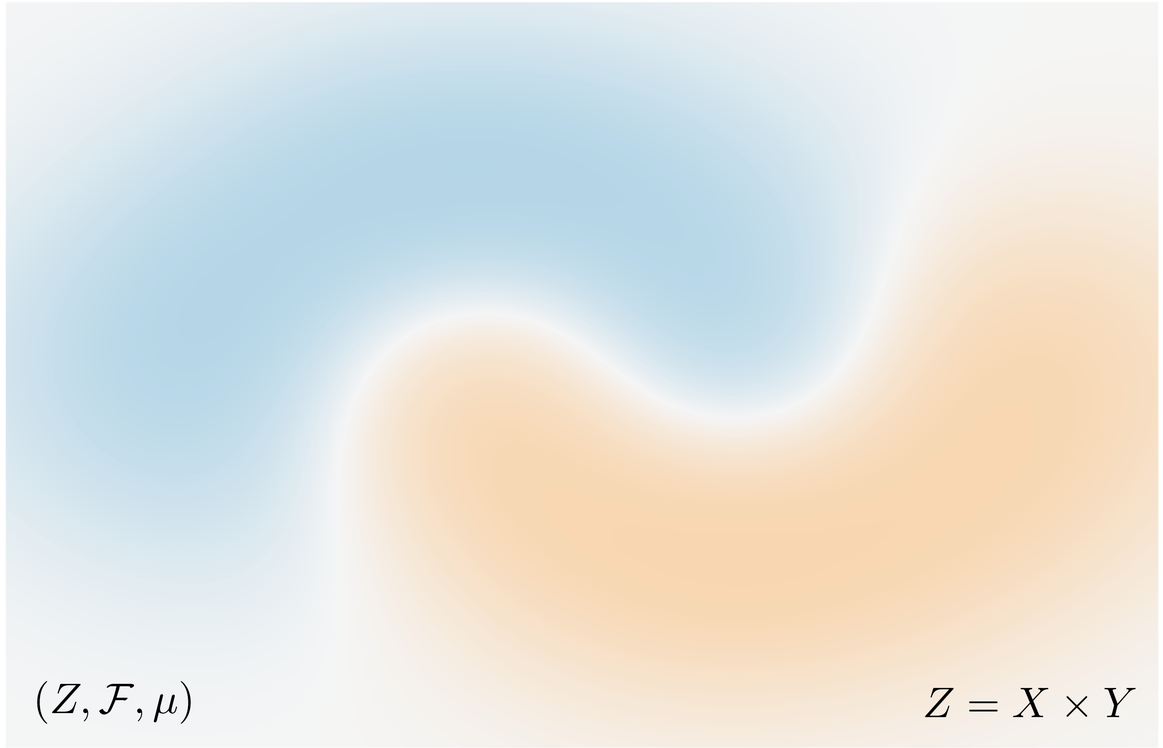
What is Deep Learning?

Session 1 — Geometric and topological methods in deep learning

The setting of supervised learning

Basic definitions

Data space

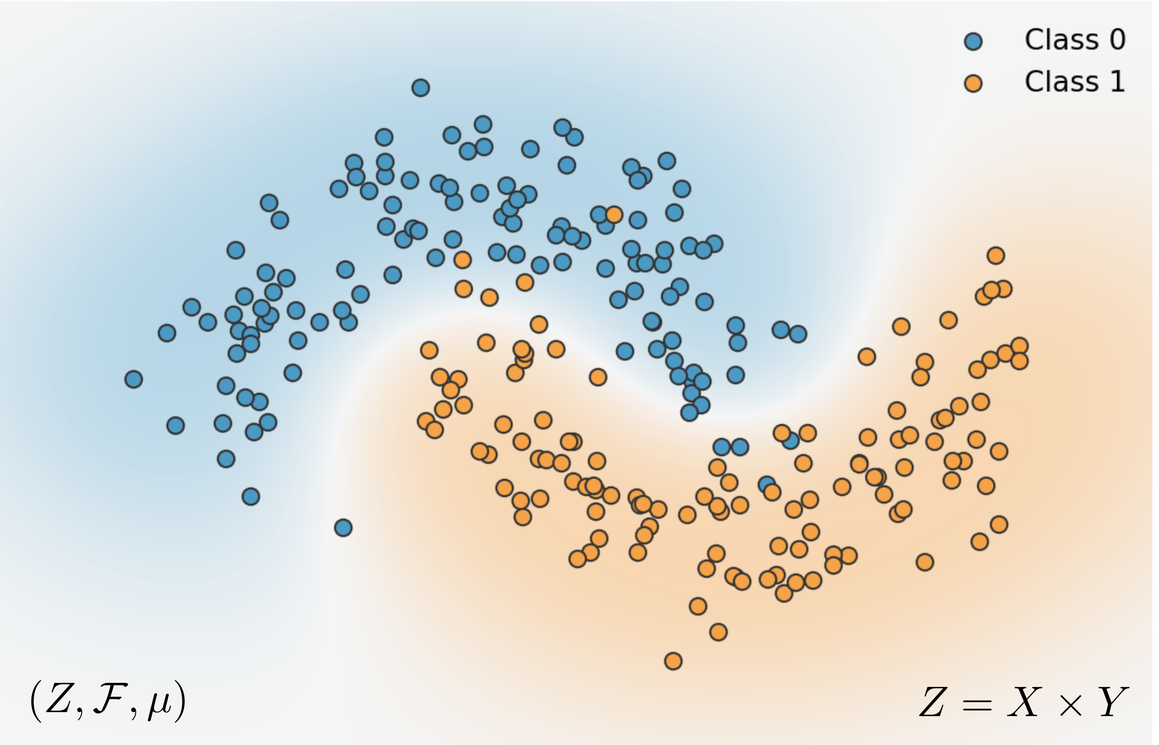


$X = \mathbb{R}^2$ $Y = \{0, 1\}$

The setting of supervised learning

Basic definitions

Training data



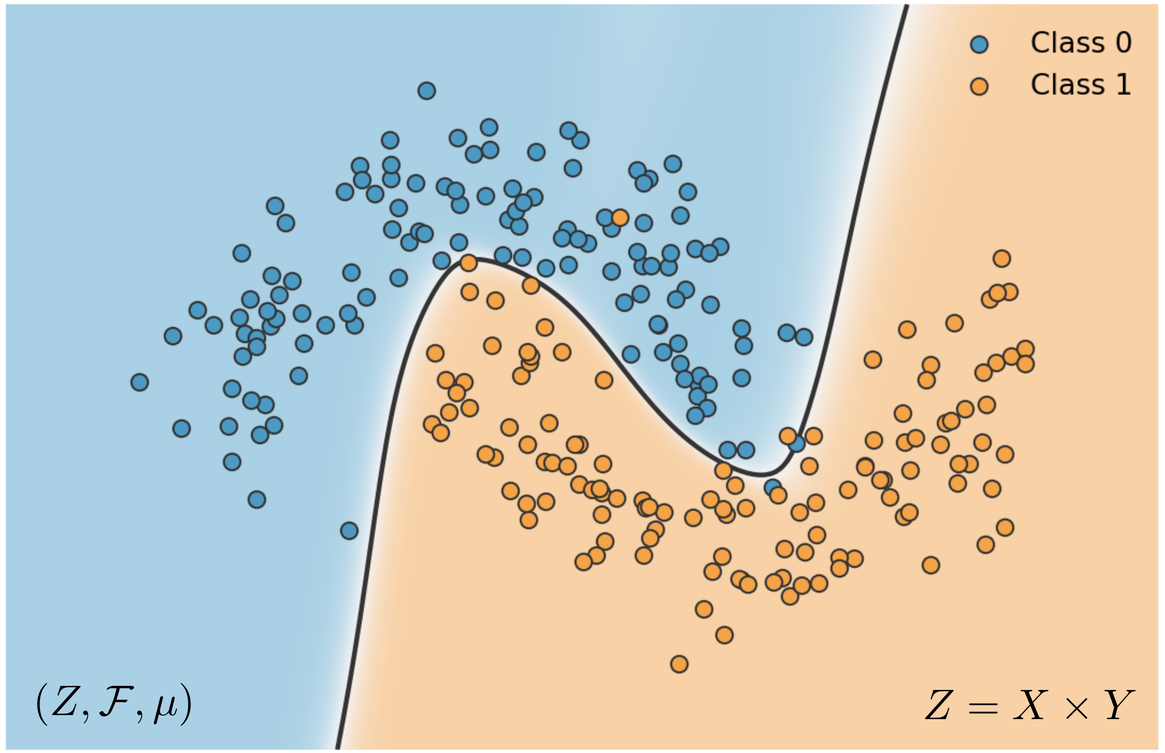
$$X = \mathbb{R}^2 \qquad Y = \{0, 1\}$$

$$\{x_1, \ldots, x_n\} \subset X \qquad \{y_1, \ldots, y_n\} \subset Y \qquad (x_i, y_i) \sim \mu$$

The setting of supervised learning

Basic definitions

Learning a parameterized approximation



$X = \mathbb{R}^2$ $\quad$ $Y = \{0, 1\}$

$\{x_1, \ldots, x_n\} \subset X$ $\quad$ $\{y_1, \ldots, y_n\} \subset Y$ $\quad$ $(x_i, y_i) \sim \mu$

Model

Learn $\hat{y} = f_\theta(x), \quad \theta \in \mathbb{R}^D$,

s.t. $\widehat{\mathcal{R}}(\theta, \{(x_i, y_i)\}) = \frac{1}{n} \sum_{i=1}^{n} \mathcal{L}(f_\theta(x_i), y_i)$ is minimized

Empirical risk

Loss

$\mathcal{L} : Z \times \mathbb{R}^D \to \mathbb{R}$

$((x, y), \theta) \mapsto \mathcal{L}(f_\theta(x), y)$

Training a neural network

Gradient descent

Input: $f_\theta : \mathbb{R}^m \to \mathbb{R}^k, \quad \mathcal{L}((x, y), \theta)$ (Training point)

1. Initialize weights randomly: θ_0
2. Loop until convergence:
   1. Compute gradient: $\dfrac{\partial \mathcal{L}}{\partial \theta} \in \mathbb{R}^D$
   2. Update weights:

Gradient over training data:
$$\theta_{t+1} \leftarrow \theta_t - \eta \frac{1}{n} \sum_{i=1}^{n} \frac{\partial \mathcal{L}}{\partial \theta}((x_i, y_i), \theta_t)$$

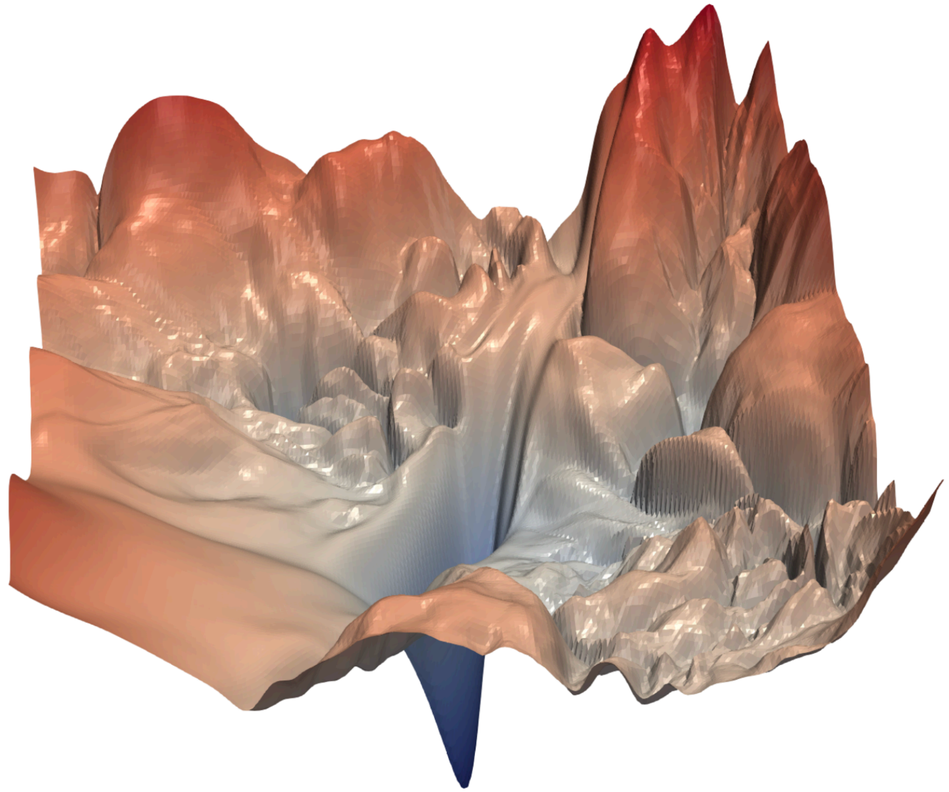
Gradient over a batch:
$$\theta_{t+1} \leftarrow \theta_t - \eta \frac{1}{|B|} \sum_{i \in B} \frac{\partial \mathcal{L}}{\partial \theta}((x_i, y_i), \theta_t)$$

(Learning rate; Batch size)

Output: Optimal weights after convergence θ_T

Collecting all the models $\{\theta_0, \theta_1, \ldots, \theta_T\}$ as we iterate our optimization algorithm we obtain a point cloud called the **optimization trajectory**

This is sample of the **loss landscape**, embedded in the parameter space $\mathbb{R}^D$



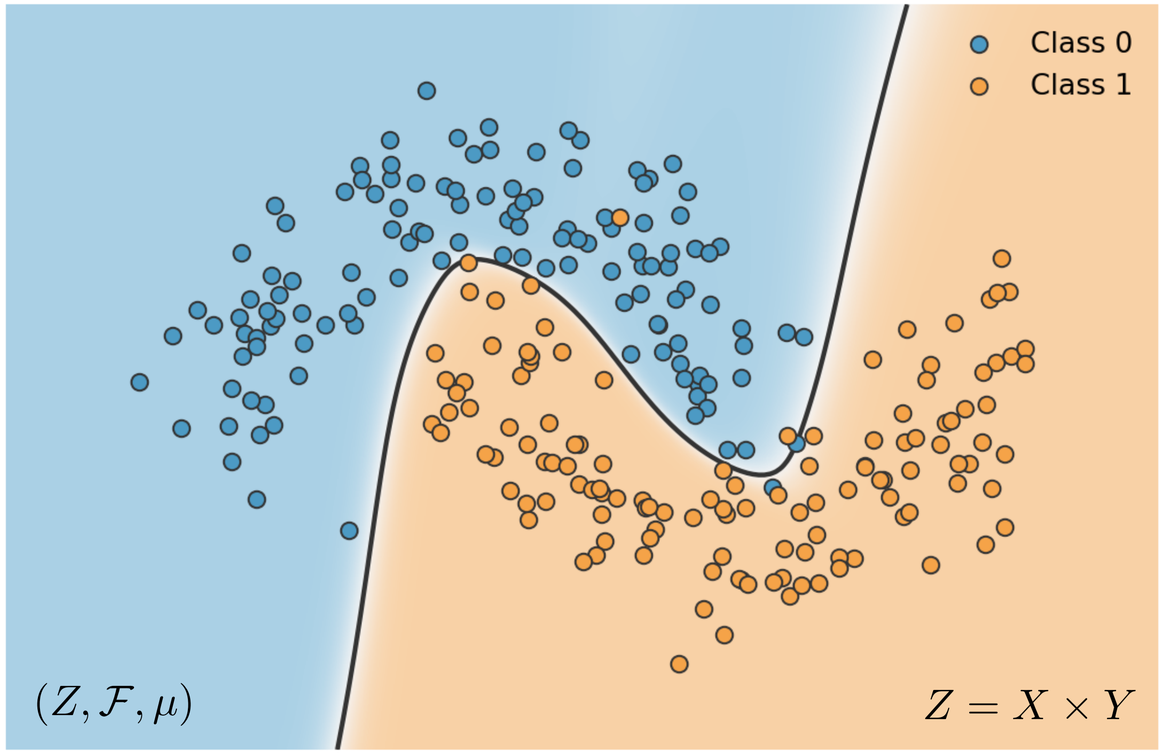
H. Li et al., Visualizing the Loss Landscape of Neural Nets, NeurIPS 2018

Generalization in neural networks

Behaviour on test data

Learn $\hat{y} = f_\theta(x), \quad \theta \in \mathbb{R}^D$,

s.t. $\widehat{\mathcal{R}}(\theta, \{(x_i, y_i)\}) = \frac{1}{n}\sum_{i=1}^{n} \mathcal{L}(f_\theta(x_i), y_i)$ is minimized

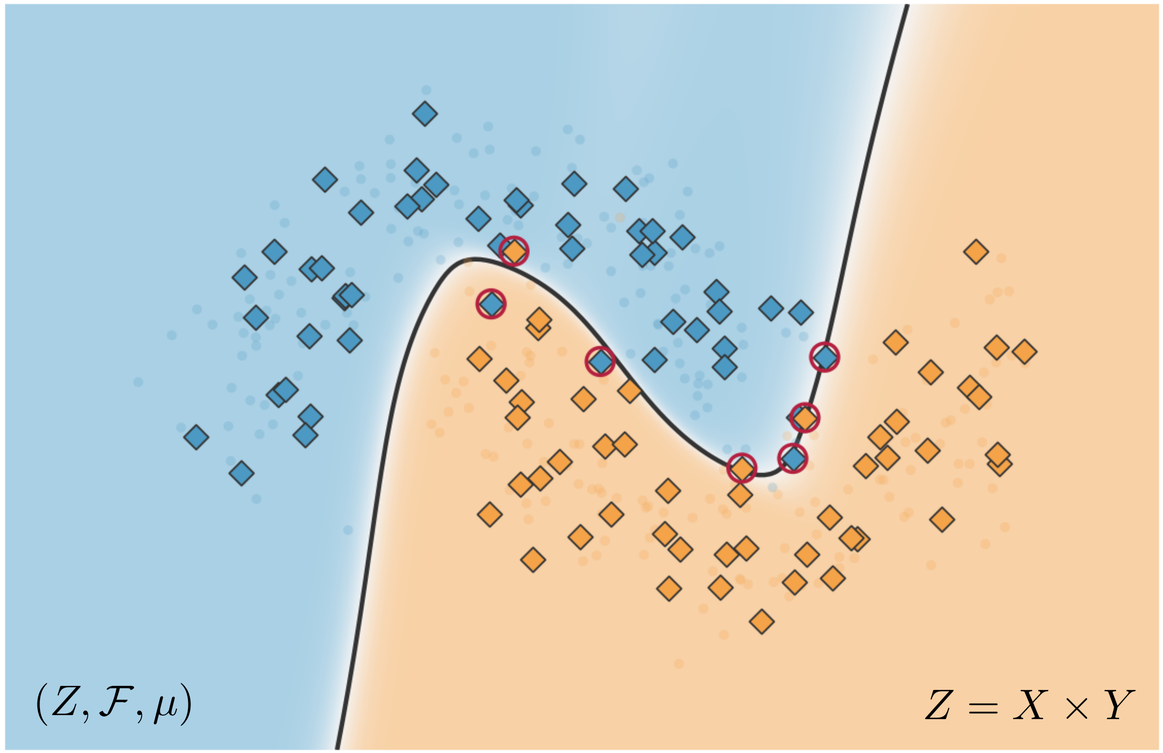


$X = \mathbb{R}^2 \qquad Y = \{0, 1\}$

$\{x_1, \ldots, x_n\} \subset X \qquad \{y_1, \ldots, y_n\} \subset Y \qquad (x_i, y_i) \sim \mu$

Generalization in neural networks

Behaviour on test data



$X = \mathbb{R}^2$ $\quad$ $Y = \{0, 1\}$

$\{x_1, \ldots, x_n\} \subset X$ $\quad$ $\{y_1, \ldots, y_n\} \subset Y$ $\quad$ $(x_i, y_i) \sim \mu$

Learn $\quad \hat{y} = f_\theta(x), \quad \theta \in \mathbb{R}^D$,

s.t. $\widehat{\mathcal{R}}(\theta, \{(x_i, y_i)\}) = \frac{1}{n} \sum_{i=1}^{n} \mathcal{L}(f_\theta(x_i), y_i)$ is minimized

We seek that the model *generalizes* well:
provides good predictions for unseen data

Population risk: $\quad \mathcal{R}(\mu, \theta) = \mathbb{E}_{z \sim \mu}[\mathcal{L}(f_\theta(x), y)]$

$$\approx \frac{1}{|S_{\text{test}}|} \sum_{(x,y) \in S_{\text{test}}} \mathcal{L}(y, f_\theta(x))$$

Typically approximated as risk over some test data

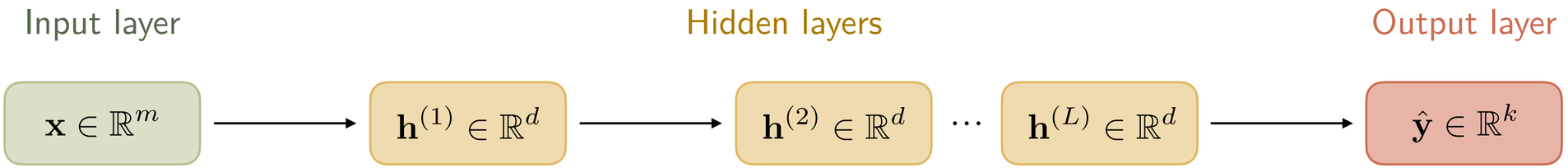
Generalization gap: $\quad \mathcal{G} = |\widehat{\mathcal{R}}(\theta, \{(x_i, y_i)\}) - \mathcal{R}(\mu, \theta)|$

Typically approximated as train - test accuracy

Neural networks

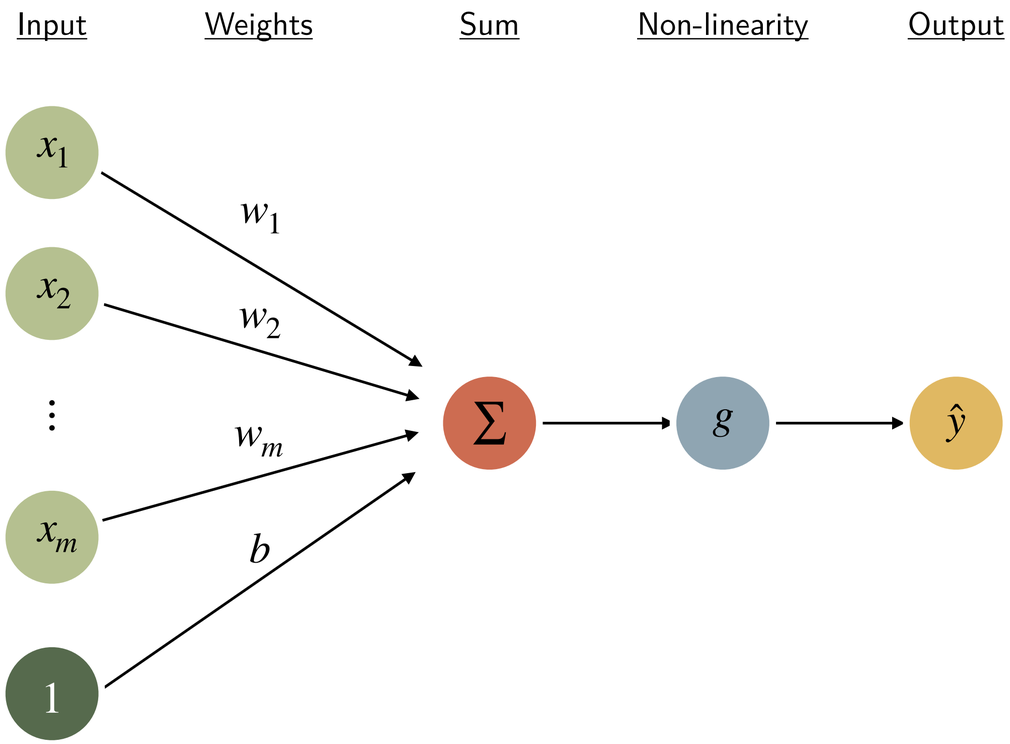
Basic definitions

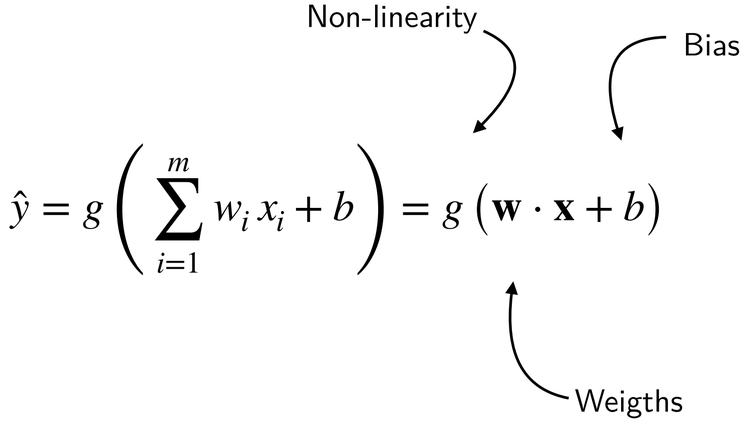
$$f_\theta : \mathbb{R}^m \to \mathbb{R}^k$$



The Perceptron

The structural building block of deep learning



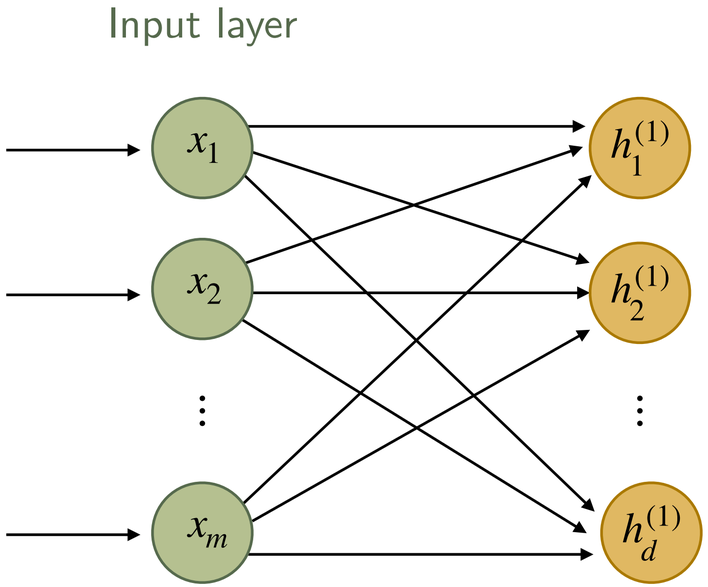


$$\hat{y} = g\left(\sum_{i=1}^{m} w_i\, x_i + b\right) = g\left(\mathbf{w}\cdot\mathbf{x} + b\right)$$

$$\mathbf{w} = (w_1, \ldots, w_m) \qquad \mathbf{x} = (x_1, \ldots, x_m)$$

The Multi-Layer Perceptron

Components



$$\mathbf{h}^{(1)} = g\left(\mathbf{W}^{(1)} \cdot \mathbf{x} + \mathbf{b}^{(1)}\right)$$

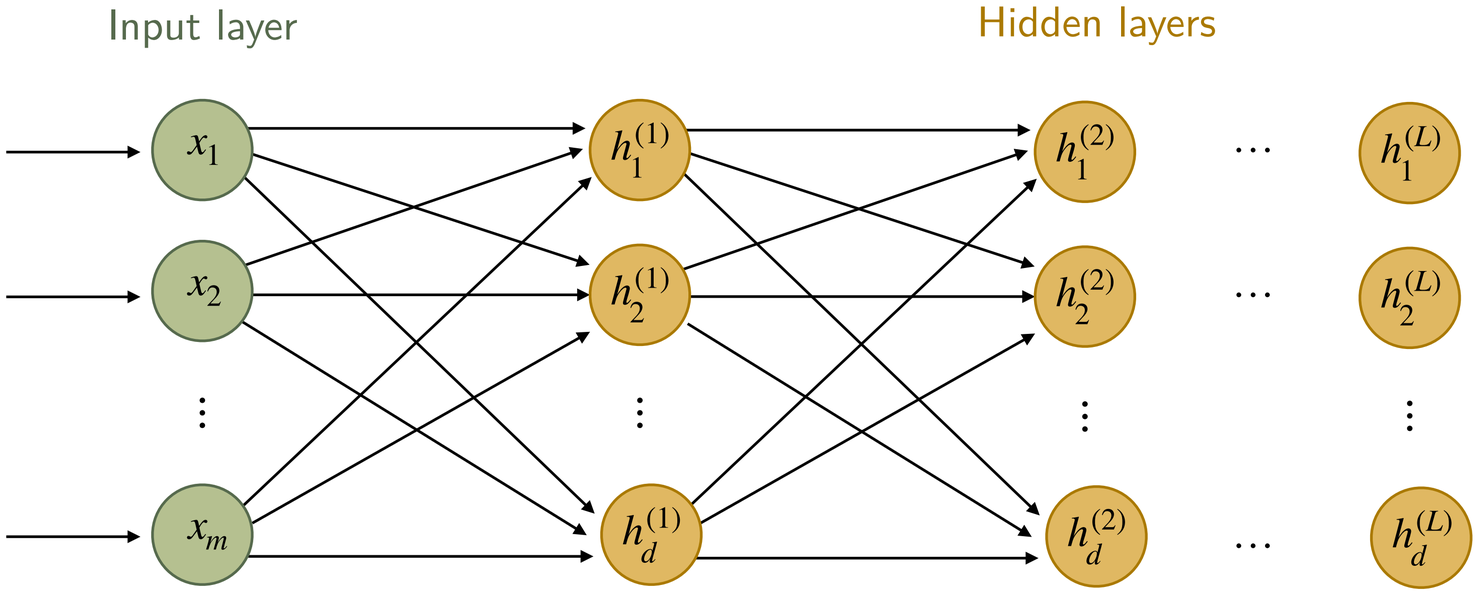
$$\mathbf{h}^{(1)} = (h_1^{(1)}, \ldots, h_d^{(1)})$$

$$\mathbf{W}^{(1)} = \begin{pmatrix} w_{1,1}^{(1)} & \cdots & w_{1,m}^{(1)} \\ \vdots & \ddots & \vdots \\ w_{d,1}^{(1)} & \cdots & w_{d,m}^{(1)} \end{pmatrix}$$

$$\mathbf{b}^{(1)} = (b_1^{(1)}, \ldots, b_d^{(1)})$$

The Multi-Layer Perceptron

Components



$$\mathbf{h}^{(1)} = g\left(\mathbf{W}^{(1)} \cdot \mathbf{x} + \mathbf{b}^{(1)}\right) \qquad \mathbf{h}^{(l)} = g\left(\mathbf{W}^{(l)} \cdot \mathbf{h}^{(l-1)} + \mathbf{b}^{(l)}\right)$$

$$\mathbf{h}^{(1)} = (h_1^{(1)}, \ldots, h_d^{(1)}) \qquad \mathbf{h}^{(l)} = (h_1^{(l)}, \ldots, h_d^{(l)})$$

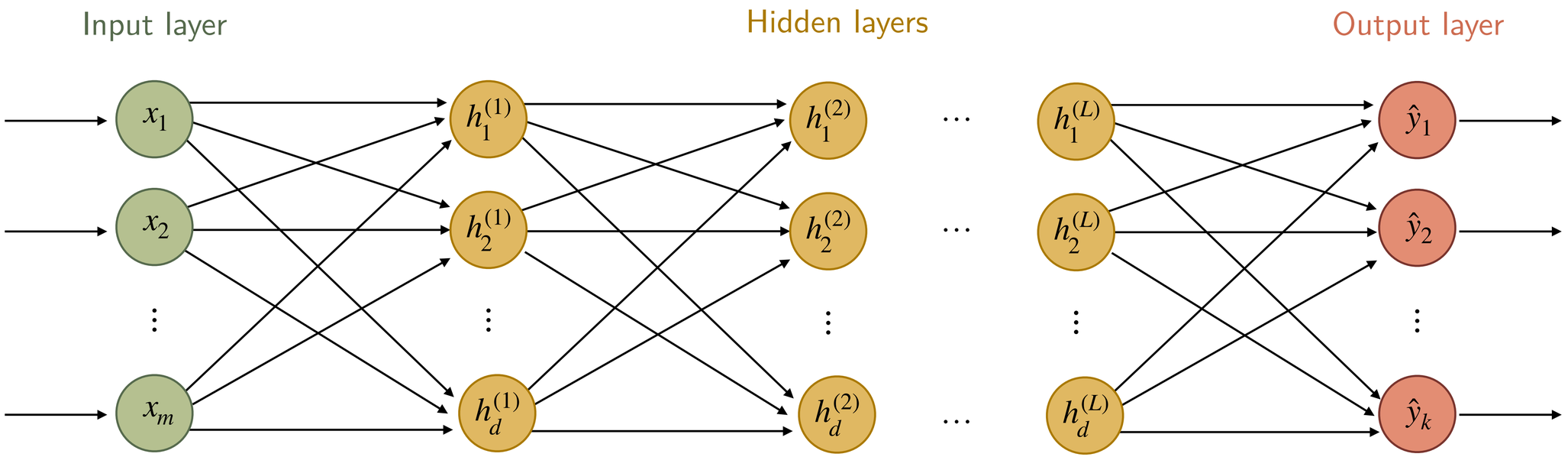
$$\mathbf{W}^{(1)} = \begin{pmatrix} w_{1,1}^{(1)} & \cdots & w_{1,m}^{(1)} \\ \vdots & \ddots & \vdots \\ w_{d,1}^{(1)} & \cdots & w_{d,m}^{(1)} \end{pmatrix} \qquad \mathbf{W}^{(l)} = \begin{pmatrix} w_{1,1}^{(l)} & \cdots & w_{1,d}^{(l)} \\ \vdots & \ddots & \vdots \\ w_{d,1}^{(l)} & \cdots & w_{d,d}^{(l)} \end{pmatrix}$$

$$\mathbf{b}^{(1)} = (b_1^{(1)}, \ldots, b_d^{(1)}) \qquad \mathbf{b}^{(l)} = (b_1^{(l)}, \ldots, b_d^{(l)})$$

The Multi-Layer Perceptron

Components

$$f_\theta : \mathbb{R}^m \to \mathbb{R}^k, \quad \theta \in \mathbb{R}^D$$



$$\mathbf{h}^{(1)} = g\left(\mathbf{W}^{(1)} \cdot \mathbf{x} + \mathbf{b}^{(1)}\right) \qquad \mathbf{h}^{(l)} = g\left(\mathbf{W}^{(l)} \cdot \mathbf{h}^{(l-1)} + \mathbf{b}^{(l)}\right) \qquad \hat{\mathbf{y}} = g\left(\mathbf{W}^{(L+1)} \cdot \mathbf{h}^{(L)} + \mathbf{b}^{(L+1)}\right)$$

$$\mathbf{h}^{(1)} = (h_1^{(1)}, \ldots, h_d^{(1)}) \qquad \mathbf{h}^{(l)} = (h_1^{(l)}, \ldots, h_d^{(l)}) \qquad \hat{\mathbf{y}} = (\hat{y}_1, \ldots, \hat{y}_k)$$

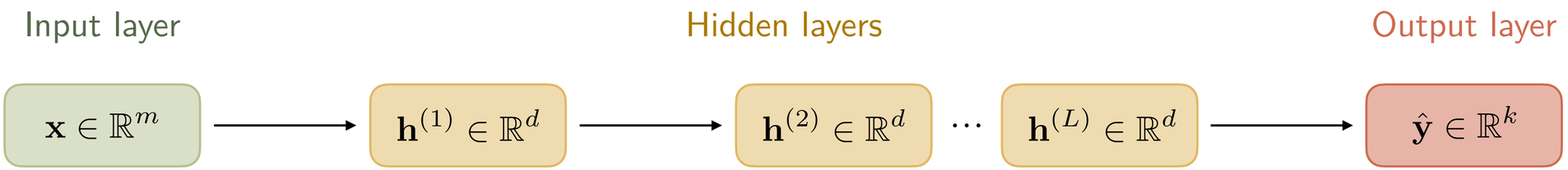
$$\mathbf{W}^{(1)} = \begin{pmatrix} w_{1,1}^{(1)} & \cdots & w_{1,m}^{(1)} \\ \vdots & \ddots & \vdots \\ w_{d,1}^{(1)} & \cdots & w_{d,m}^{(1)} \end{pmatrix} \qquad \mathbf{W}^{(l)} = \begin{pmatrix} w_{1,1}^{(l)} & \cdots & w_{1,d}^{(l)} \\ \vdots & \ddots & \vdots \\ w_{d,1}^{(l)} & \cdots & w_{d,d}^{(l)} \end{pmatrix} \qquad \mathbf{W}^{(L+1)} = \begin{pmatrix} w_{1,1}^{(L+1)} & \cdots & w_{1,d}^{(L+1)} \\ \vdots & \ddots & \vdots \\ w_{k,1}^{(L+1)} & \cdots & w_{k,d}^{(L+1)} \end{pmatrix}$$

$$\mathbf{b}^{(1)} = (b_1^{(1)}, \ldots, b_d^{(1)}) \qquad \mathbf{b}^{(l)} = (b_1^{(l)}, \ldots, b_d^{(l)}) \qquad \mathbf{b}^{(L+1)} = (b_1^{(L+1)}, \ldots, b_k^{(L+1)})$$

Neural networks

Basic definitions

$$f_\theta : \mathbb{R}^m \to \mathbb{R}^k$$



Convolutional neural networks

Autoencoders

Graph neural networks

Generative Adversarial Networks

Recurring neural networks

Transformers

Motivation

How to capture shape features?

Suppose you are given the following point cloud, how can you design a test to conclude that this sample comes from a distribution over *two circles*?



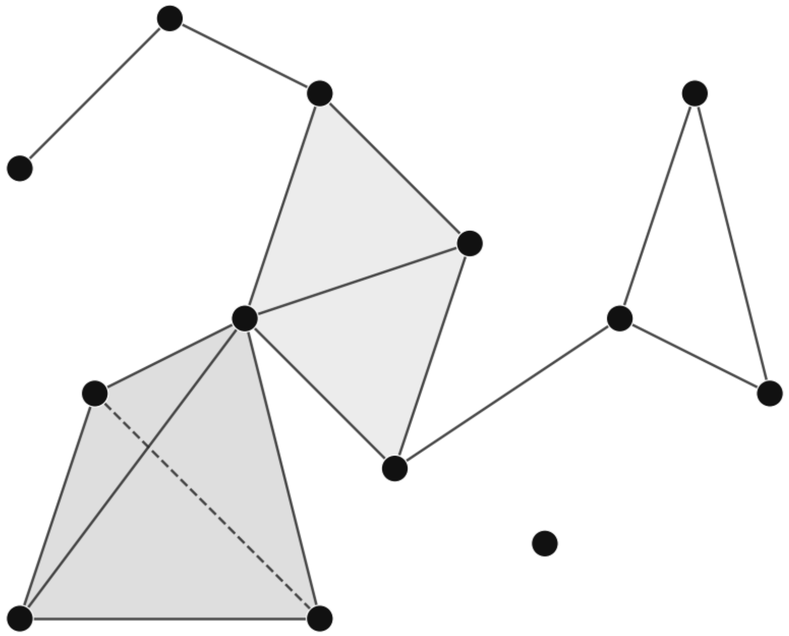
We first need to approximate the point cloud with some geometric object

Simplicial Homology

Counting holes in simplicial complexes

Simplicial complexes

Definition



Def. (Abstract) Simplicial Complex

Given a discrete set $S = \{s_1, \ldots, s_n\}$, a simplicial complex X is a set of subsets $\sigma \subset S$ *closed under inclusion* (if $\sigma \in X$, then for all $\tau \subseteq \sigma$ we must have $\tau \in X$)

We can *stratify* a simplicial complex by dimension:

Vertices: 0-dimensional simplices

Edges: 1-dimensional simplices

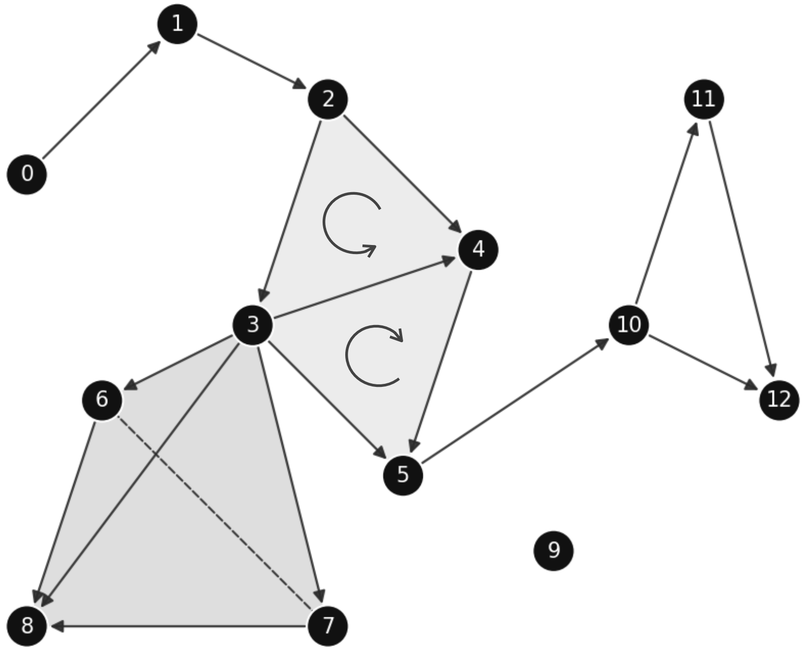
Triangles: 2-dimensional simplices

In general:

- k-**dimensional simplex**: $\sigma = \{s_{i_0}, \ldots, s_{i_k}\}$
- X_k: set of k-dimensional simplices
- k-**skeleton** $(X_{\leq k})$: all simplices with dimension $\leq k$

Simplicial homology

Chain spaces



The first thing we do is choosing an **orientation** for the simplices, that is, a unique way of representing a set $\sigma = \{s_{i_0}, \ldots, s_{i_k}\}$, by *ordering the nodes* and ordering the simplices accordingly: $(s_{i_0}, \ldots, s_{i_k}) \equiv (i_0, \ldots, i_k)$, $i_0 < \ldots < i_k$

Chain space: let $k \geq 0$, the k-chain space is the *vector space* spanned by the *oriented simplices*

$$C_k(X) = \text{span}\{(i_0, \ldots, i_k) : i_0 < \cdots < i_k\} \cong \mathbb{R}^{|X_k|}$$

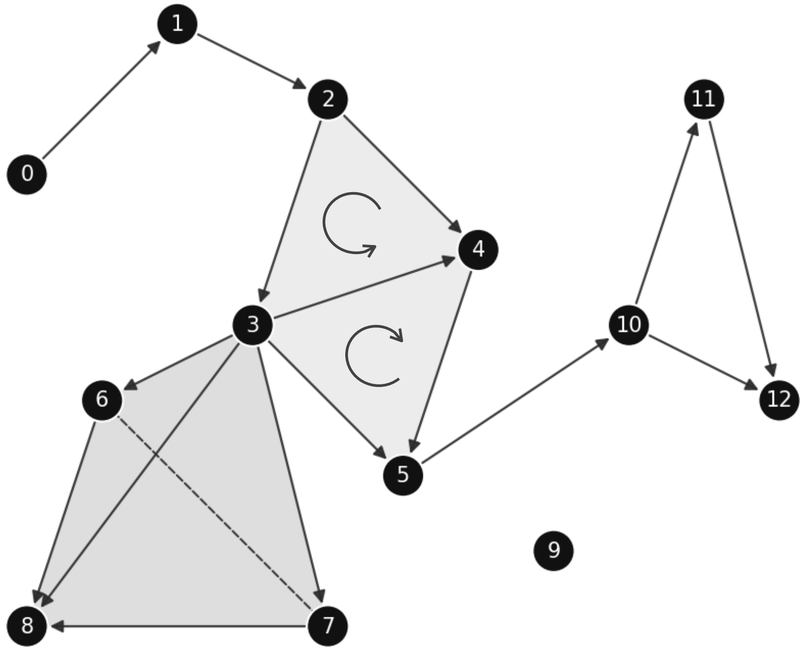
$$c = \sum_{\sigma \in X_k} a_\sigma \cdot \sigma, \quad a_\sigma \in \mathbb{R}$$

Example:

$$C_0(X, \mathbb{R}) = \mathbb{R}^{13} \qquad C_1(X, \mathbb{R}) = \mathbb{R}^{17} \qquad C_2(X, \mathbb{R}) = \mathbb{R}^6 \qquad C_3(X, \mathbb{R}) = \mathbb{R}$$

Simplicial homology

Boundary operators



Boundary operator: *linear map* connecting chain spaces of consecutive dimension $\partial_k : C_k(X) \to C_{k-1}(X)$ defined on the basis elements as

$$\partial_k(i_0, \dots, i_k) = \sum_{j=0}^{k} (-1)^j \ (i_0, \dots, \widehat{i_j}, \dots i_k)$$

And extended linearly:

$$c = \sum_{i_0 < \cdots < i_k} a_{(i_0,\dots,i_k)} \cdot (i_0, \dots, i_k), \quad a_\sigma \in \mathbb{R}$$

$$\partial_k c = \sum_{i_0 < \cdots < i_k} a_{(i_0,\dots,i_k)} \cdot \partial_k(i_0, \dots, i_k)$$

Example:

$$\partial_1(0,1) = (1) - (0)$$
$$\partial_2(2,3,4) = (3,4) - (2,4) + (2,3)$$
$$\partial_3(3,6,7,8) = (6,7,8) - (3,7,8) + (3,6,8) - (3,6,7)$$

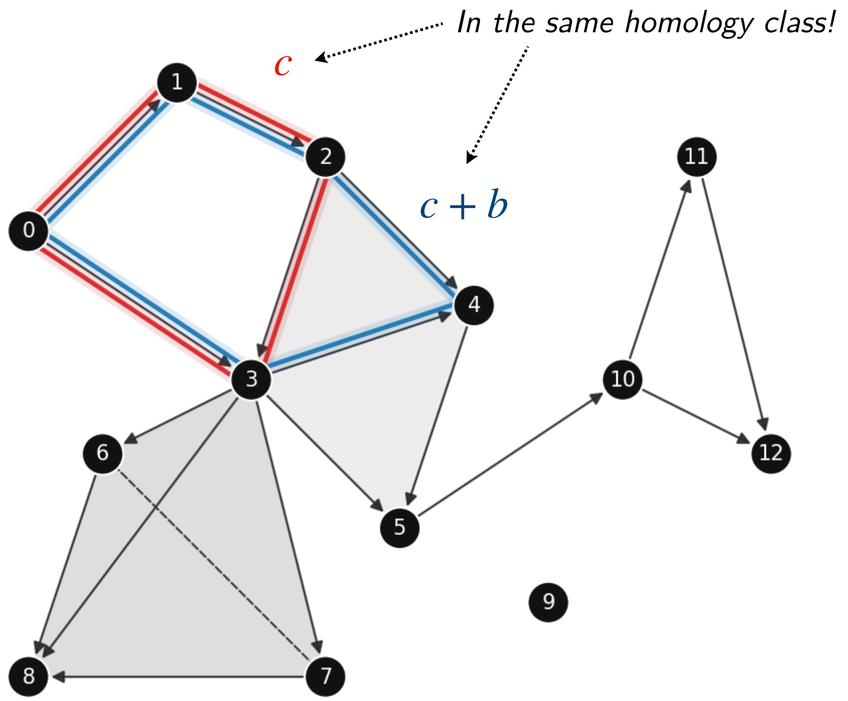
Important property:

$$\partial_k \, \partial_{k+1} = 0 \implies \ker \partial_k \subseteq \operatorname{im} \partial_{k+1}$$

Cycles *Boundaries*

Simplicial homology

Homology groups and Betti numbers



If I can move through filled in triangles, I stay inside of the homology class

Def. Homology group

Cycles

$$H_k(X) = \frac{\ker \partial_k}{\operatorname{im} \partial_{k+1}}$$

Boundaries

Def. Betti number

$$\beta_k(X) = \dim H_k(X)$$

"Number of k-dimensional holes in X"

- $k = 0$: number of connected components
- $k = 1$: number of loops
- $k = 2$: number of cavities or bubles

Example: $\beta_0(X) = 2$ $\beta_1(X) = 2$

$\beta_2(X) = 1$ $\beta_k(X) = 0,\ k \geq 3$

Simplicial homology

Functoriality

Def. Simplicial map

A simplicial map $f : X \to Y$ is a *map between the sets of vertices* of the simplicial complexes such that for every simplex $\sigma = \{x_0, \ldots, x_k\} \in X$, its image $f(\sigma) = \{f(x_0), \ldots, f(x_k)\} \in Y$ is also a simplex

Given a simplicial map $f : X \to Y$, we can define two induced maps:

- For **chains**: $f_\# : C_k(X) \to C_k(Y)$ by sending $\sigma \mapsto f(\sigma)$ if $f(\sigma) \in Y_k$, and 0 otherwise; and extend linearly
- This assignment respects boundaries: $\partial_k \circ f_\# = f_\# \circ \partial_k$

Hence, we also have a linear map for **homology groups**, $f_* : H_k(X) \to H_k(Y)$ which in addition, satisfies:

$$(f \circ g)_* = f_* \circ g_* \qquad\qquad (\mathrm{id}_X)_* = \mathrm{id}_{H_k(X)}$$

In other words, the assignment $H_k(-) : \mathrm{Simp} \to \mathrm{Vect}$, $X \mapsto H_k(X)$ is a *covariant* functor

Approximating point clouds with simplicial complexes

Measuring the shape of point clouds

Approximating point clouds using simplicial complexes

Čech complex

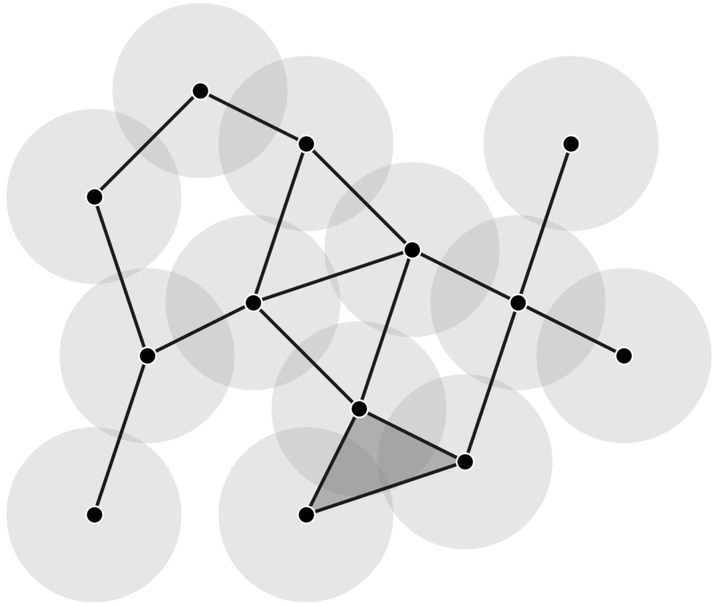
Finite metric space

Given (S, d) and $r \geq 0$, the **Čech complex** (or Nerve complex) at scale r is:

$$C_r(S, d) = \left\{ \{s_{i_0}, \ldots, s_{i_k}\} : \bigcap_{j=0}^{k} B(s_{i_j}, r) \neq \varnothing \right\}$$

Monotonicity

For $r \leq s$: $C_r(S) \subseteq C_s(S)$



Approximating point clouds using simplicial complexes

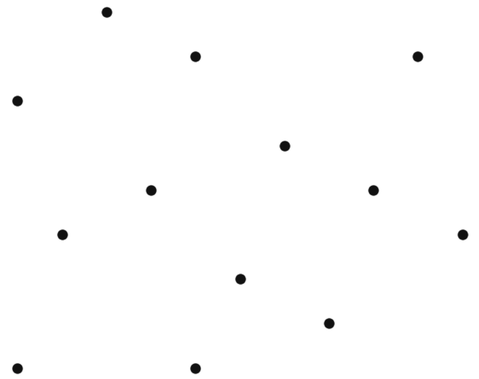
Vietoris—Rips

Given (S, d) and $r \geq 0$, the **Vietoris-Rips complex** at scale r is:

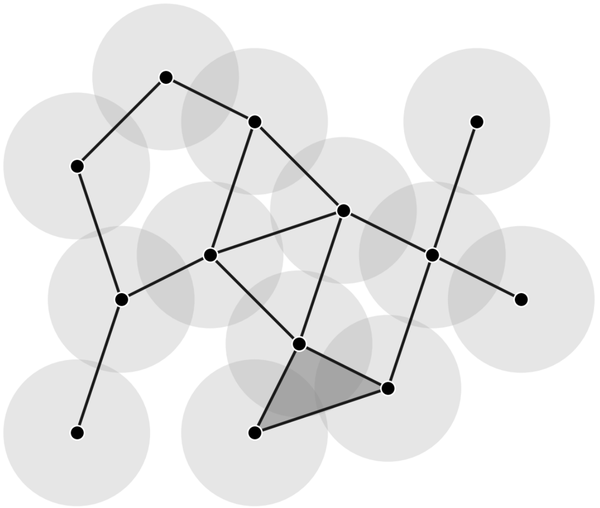
$$VR_r(S, d) = \left\{ \{s_{i_0}, \ldots, s_{i_k}\} : d(s_i, s_j) \leq 2r,\ \forall i_0 \leq i < j \leq i_k \right\}$$

Monotonicity

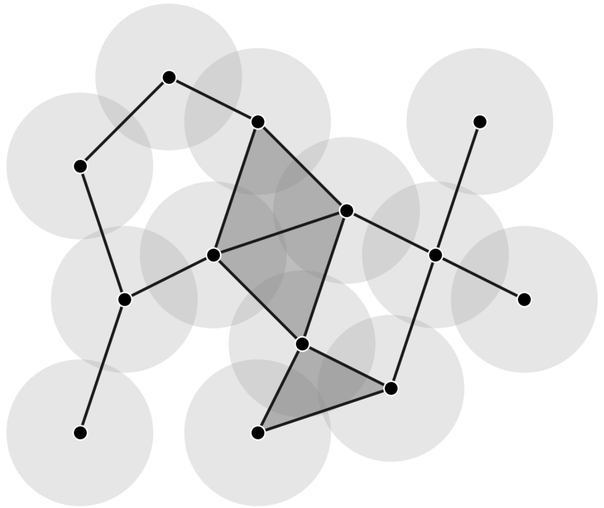
For $r \leq s$: $VR_r(S) \subseteq VR_s(S)$



Point cloud



Čech complex



VR complex

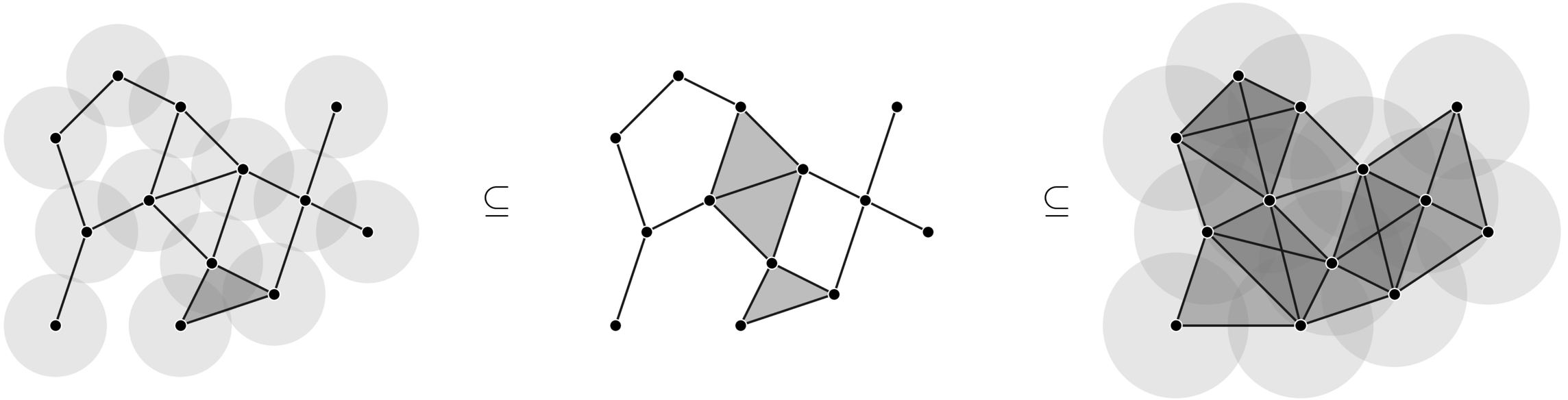
Simplicial complexes from point clouds

Relating Čech and VR complexes

Lemma:

For $r \geq 0$: $\quad C_r(S) \subseteq VR_r(S) \subseteq C_{\sqrt{2}r}(S)$

$C_r(S)$ $\quad VR_r(S)$ $\quad C_{\sqrt{2}r}(S)$



Nerve theorem

Recovering the homotopy type of a topological space from a sample

Def + thm. Nerve theorem (informal)

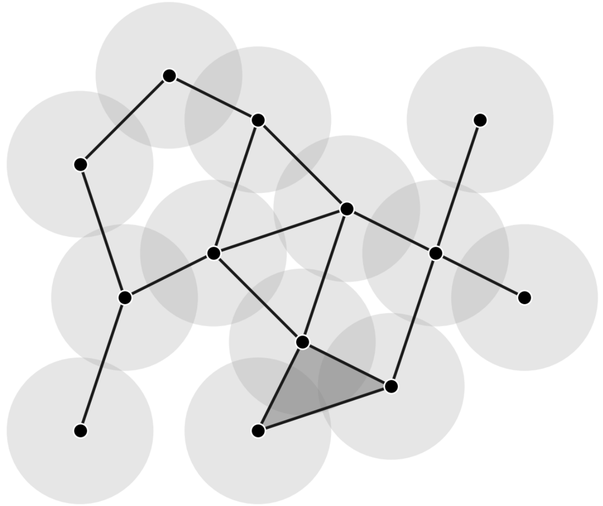
Let X be a topological space and $\mathcal{U} = \{U_i : i \in I\}$ a cover of this space: $X \subset \bigcup_{i \in I} U_i$

The *nerve* of this cover is the following simplicial complex:

$$\mathrm{Nerve}(\mathcal{U}) = \left\{ J \subseteq I : \bigcap_{j \in J} U_i \neq \emptyset \right\}$$

If any $\bigcap_{j \in J \subseteq I} U_j \neq \emptyset$ is contractible, with some *tameness conditions* on X and $\mathcal{U}$ we have:

$$\mathrm{Nerve}(\mathcal{U}) \cong X \qquad \textbf{(homotopy equivalent)}$$



The *Čech complex* is a nerve

$$C_r(S, d) = \left\{ \{s_{i_0}, \ldots, s_{i_k}\} : \bigcap_{j=0}^{k} B(s_{i_j}, r) \neq \varnothing \right\}$$

The *Vietoris-Rips complex,* although not a nerve, is isomorphic to one[1]

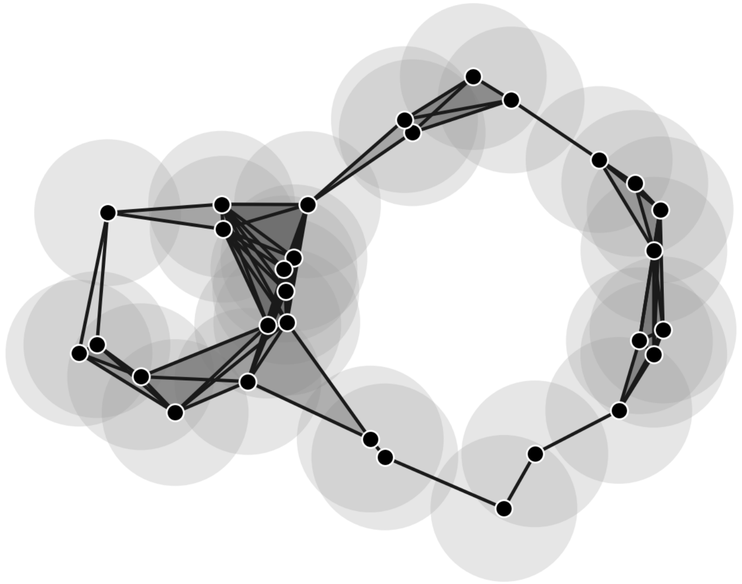
[1]Ghrist, R., and Muhammad, A. Coverage and Hole-Detection in Sensor Networks via Homology. *IPSN 2005. Fourth International Symposium on Information Processing in Sensor Networks*, 2005., April 2005, 254–60.

[1]Bauer, U., Kerber, M., Roll, F., and Rolle, A. A Unified View on the Functorial Nerve Theorem and Its Variations. *Expositiones Mathematicae* 41, no. 4 (2023): 125503.

Putting everything together

Detecting the two circles in my point cloud

1. Approximate S with a Čech complex for a some radius that captures the holes: $C_r(S)$
2. Compute the Betti numbers in dimension 1: $\beta_1\big(C_r(S)\big)$
3. If $\beta_1(C_r(S)) = 2$, then you are observing two holes



How do you find the radius r?

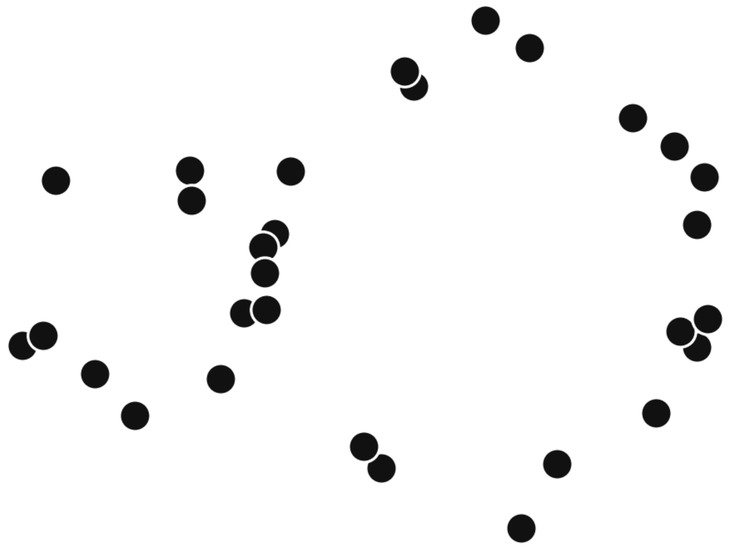
Persistent Homology

Measuring the shape of point clouds

Persistent homology

Filtrations

Consider all possible radii!



Def. Filtration

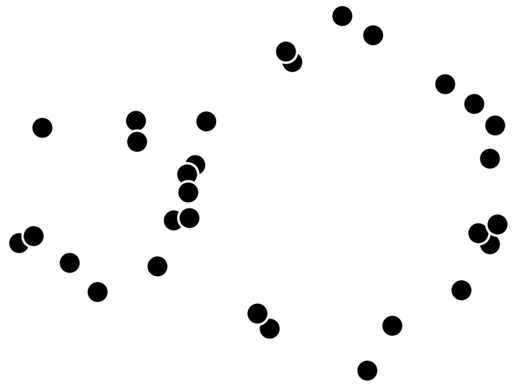
$\{X_t : t \in \mathbb{R}\}$ such that for $s \leq t$, $X_s \subseteq X_t$

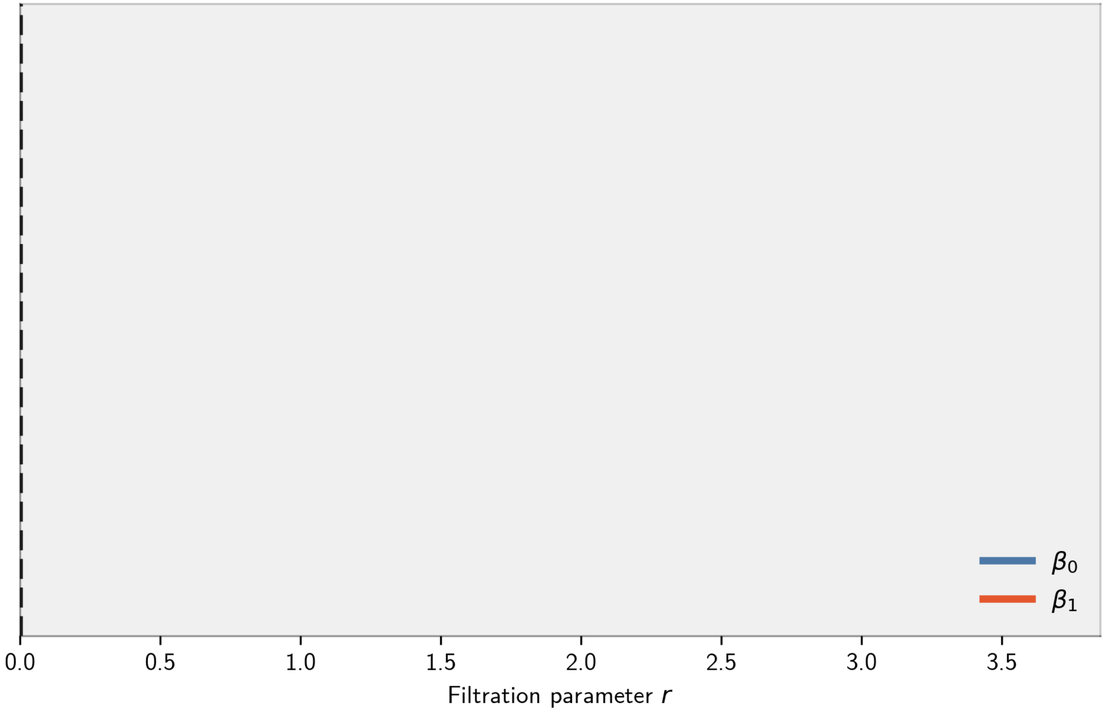
Persistent homology

Persistence barcodes

Compute the Betti numbers at each step in the filtration

Track when the Betti numbers increase or decrease and store this information in a **barcode**

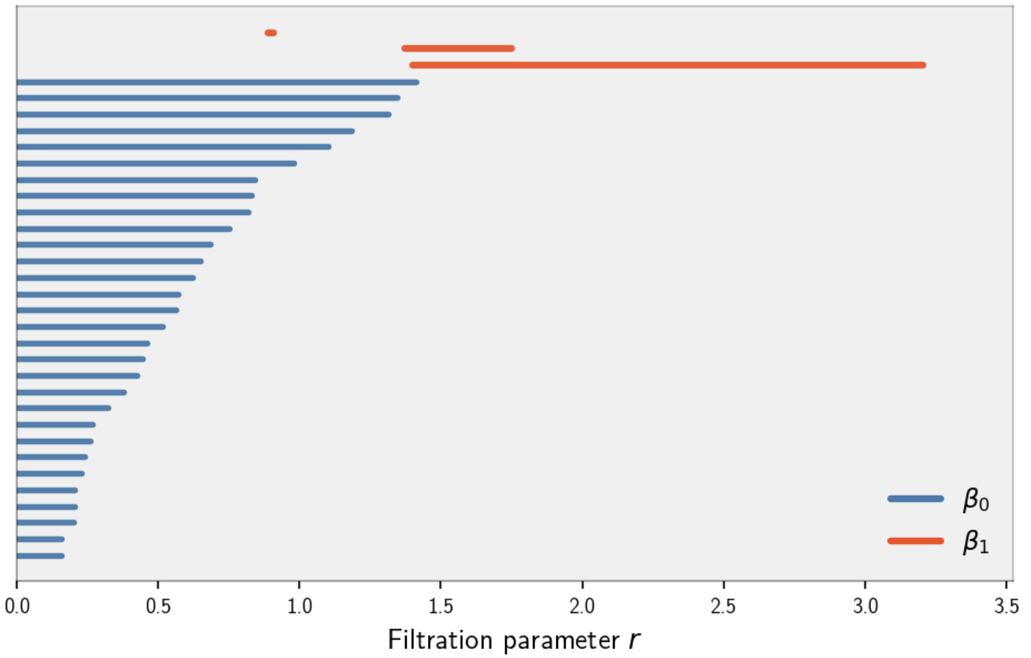




Persistent homology

Persistence barcodes

As many bars as points, bars *die* as points get connected



Two long bars: we have two *persistent* loops in the point cloud

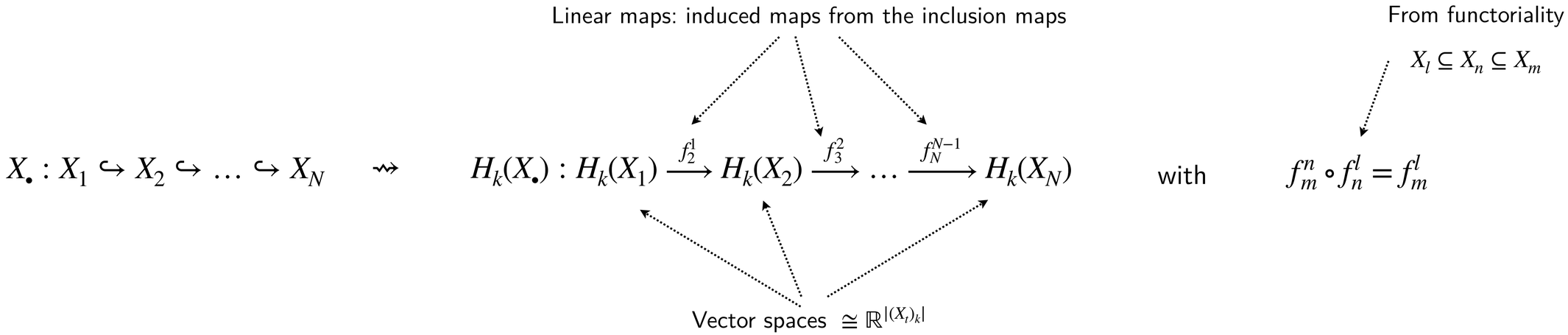
The point cloud had the shape of two circles!

Persistence barcode

Persistent homology

Persistence modules

Applying the homology functor to the filtration we arrive to the **persistent homology module**



Def. Persistence module

A family $\{V_n\}_{1 \leq n \leq N}$ of vector spaces and a family $\{\phi_m^n : V_n \to V_m\}_{n \leq m}$ of linear maps satisfying:

- $\phi_n^n = \mathrm{id}_{V_n}$
- $\phi_n^m \circ \phi_m^l = \phi_n^l$ for all $l \leq m \leq n$

Persistent homology

Persistence modules as functors

Partially ordered set or poset: $(P, \leq)$ is given by a set P with an order relation $\leq$ which is

- *Reflexive*: $p \leq p$, $\forall p \in P$
- *Antisymmetric*: $p \leq q$ and $q \leq p \implies p = q, \quad \forall p, q \in P$
- *Transitive*: $p \leq q$ and $q \leq r \implies p \leq r$

A poset can be seen as a *small category* where objects are the elements in P and there is an arrow between two $p, q \in P$ iff $p \leq q$

Small category defined by a poset

Category of vector spaces

Def. Persistence module as a functor $M : (P, \leq) \to \mathrm{Vec}_{\mathbb{k}}$

- For every $t \in P$, we have a vector space $M_t = M(t)$
- For every pair of comparable points $s \leq t$ we have a map $M(s \leq t) : M_s \to M_t$ (**transition map**)

Persistent homology

Intervals in posets

Def. Interval

An interval $I \subseteq P$ in a poset is a subset satisfying the two conditions:

- **Convexity:** for every pair $p \leq r \in I$ and any $p \leq q \leq r$, we have $q \in I$
- **Connectivity**: for every pair $p, q \in I$ there exists a sequence of comparable elements $\{p = u_0, u_1, \ldots, u_{n-1}, u_n = q\}$ with $u_i \leq u_{i+1}$ or $u_{i+1} \leq u_i$ for all i

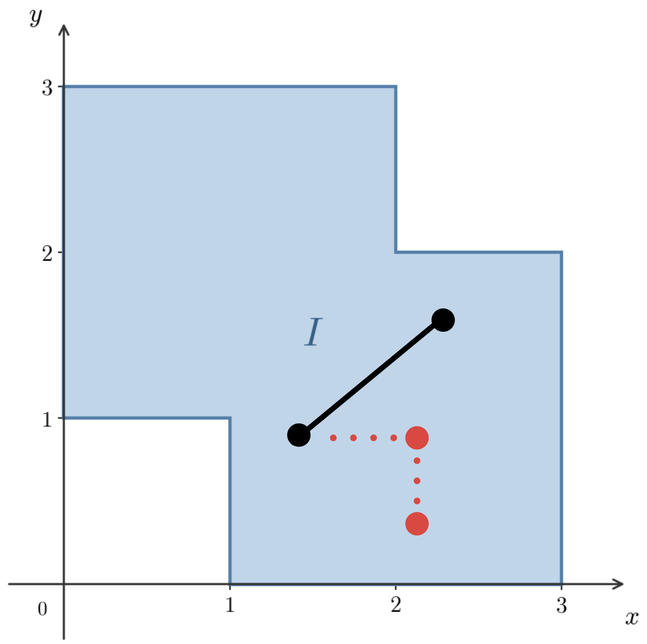
Def. Interval module

$$\Bbbk(I) : (P, \leq) \to \mathrm{Vect}$$

$$\Bbbk(I)_t = \begin{cases} \Bbbk & \text{if } t \in I \\ 0 & \text{otherwise} \end{cases} \qquad \Bbbk(I)_{t \leq s} = \begin{cases} \mathrm{id}_{\Bbbk} & \text{if } t, s \in I \\ 0 & \text{otherwise} \end{cases}$$

Example: in $(\mathbb{R}, \leq)$ this precisely recovers the intervals of the real line

Example: $(\mathbb{R}^2, \preceq)$, $(x_1, y_1) \preceq (x_2, y_2) \iff x_1 \leq x_2,\ y_1 \leq y_2$



Persistent homology

Structure theorem and barcodes

Can be extended to totally ordered posets!

Important that the target category is Vec

Thm. Structure theorem (informal)

If a persistence module $M : (\mathbb{R}, \leq) \to \mathrm{Vec}$ satisfies some tameness conditions, then it admits an *essentially unique* decomposition in interval modules:

$$M \cong \bigoplus_{i \in I} \mathbb{k}[b_i, d_i)$$

The barcode is precisely the multisite of intervals appearing in this decomposition

This justifies that it is a *complete invariant*

$$M_1 \cong M_2 \iff \mathrm{Barc}(M_1) = \mathrm{Barc}(M_2)$$

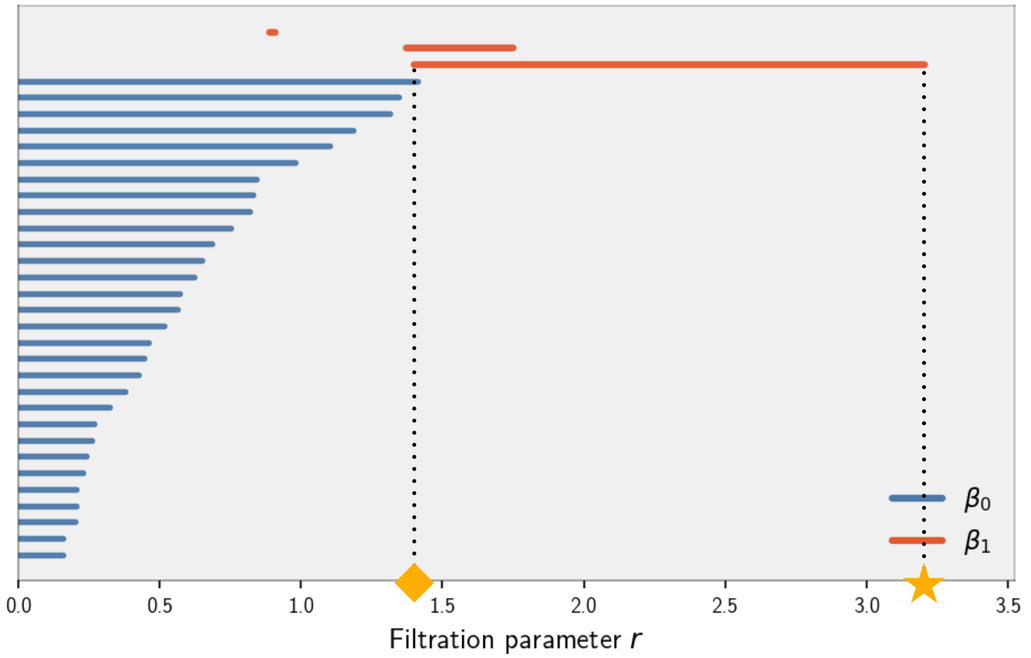
Azumaya, Gorô. Corrections and Supplementaries to My Paper Concerning Krull-Remak-Schmidt's Theorem. *Nagoya Mathematical Journal* 1 (June 1950): 117–24.

Crawley-Boevey, William. Decomposition of Pointwise Finite-Dimensional Persistence Modules. *Journal of Algebra and Its Applications* 14, no. 05 (2015): 1550066.

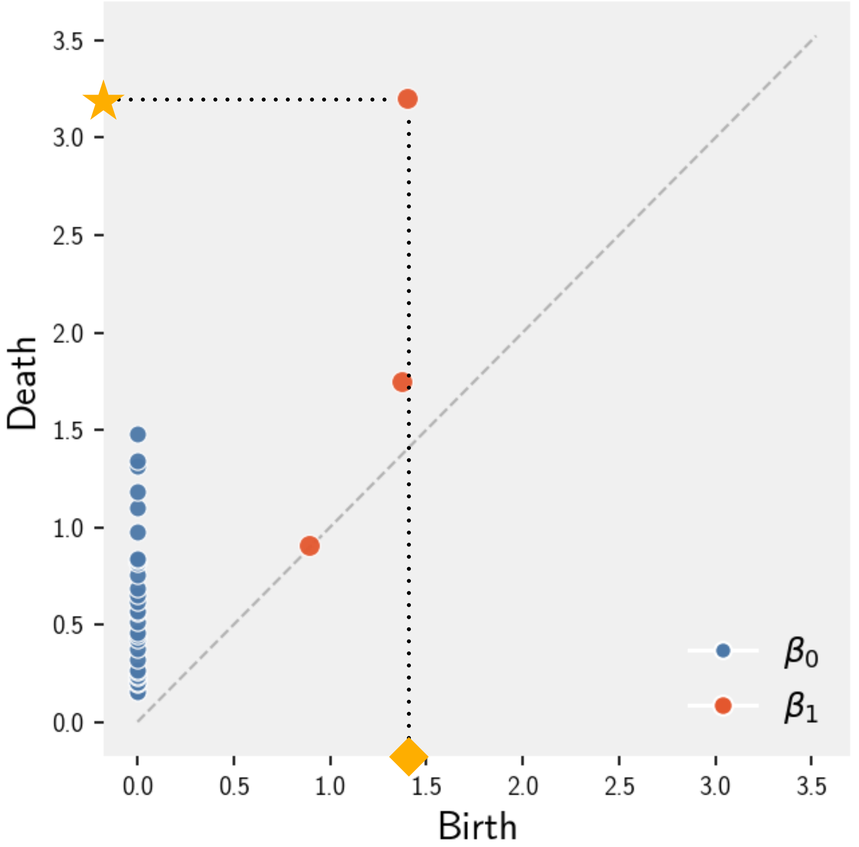
Persistent homology

Barcodes and diagrams

Persistence barcode



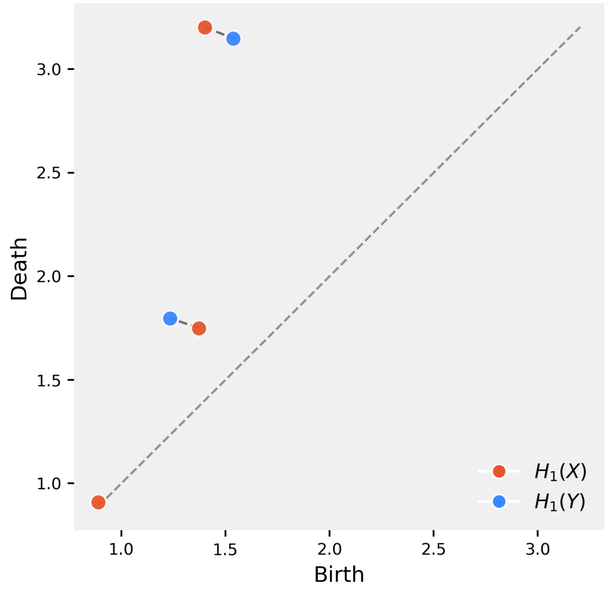
Persistence diagram



Persistent homology

Distances between persistence diagrams

Let $\psi : \mathrm{PD}(X) \to \mathrm{PD}(Y)$ denote a bijection between the points in the diagrams, where the points in the diagonal in both diagrams are counted with infinite multiplicity



$$\mathrm{PD}(X) := \mathrm{PD}(H_1(X))$$

$$\mathrm{PD}(Y) := \mathrm{PD}(H_1(Y))$$

Bottleneck distance

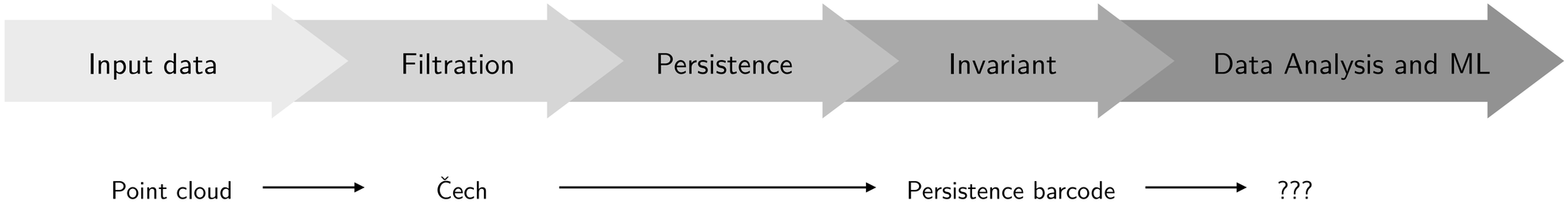
$$d_B\left(\mathrm{PD}(X), \mathrm{PD}(Y)\right) = \inf_{\psi:\mathrm{PD}(X)\to\mathrm{PD}(Y)} \max\{\|x - \psi(x)\|_\infty : \forall x \in \mathrm{PD}(X)\}$$

Wasserstein distance

$$W_p\left(\mathrm{PD}(X), \mathrm{PD}(Y)\right) = \inf_{\psi:\mathrm{PD}(X)\to\mathrm{PD}(Y)} \left(\sum_{x\in\mathrm{PD}(X)} \|x - \psi(x)\|_p^p \right)^{1/p}$$

Persistent homology

The general pipeline



The space of barcodes with W_2 is a Polish space, **amenable to statistics and probability**[1]

However, it is a space with **complicated geometry**, which hinders the development of statistical theory[2]

In addition, most ML/DL methods expect **input data** coming from an **Euclidean space**

A possible solution: *vectorize*[3] persistence diagrams, i.e., embed them in a Hilbert space where we can develop statistical theory

[1]Mileyko, Y., Mukherjee, S., and Harer, J. Probability Measures on the Space of Persistence Diagrams. *Inverse Problems* 27, no. 12 (2011): 124007.

[2]Turner, K., Mileyko, Y., Mukherjee, S., and Harer, J. Fréchet Means for Distributions of Persistence Diagrams. *Discrete & Computational Geometry* 52, no. 1 (2014): 44–70.

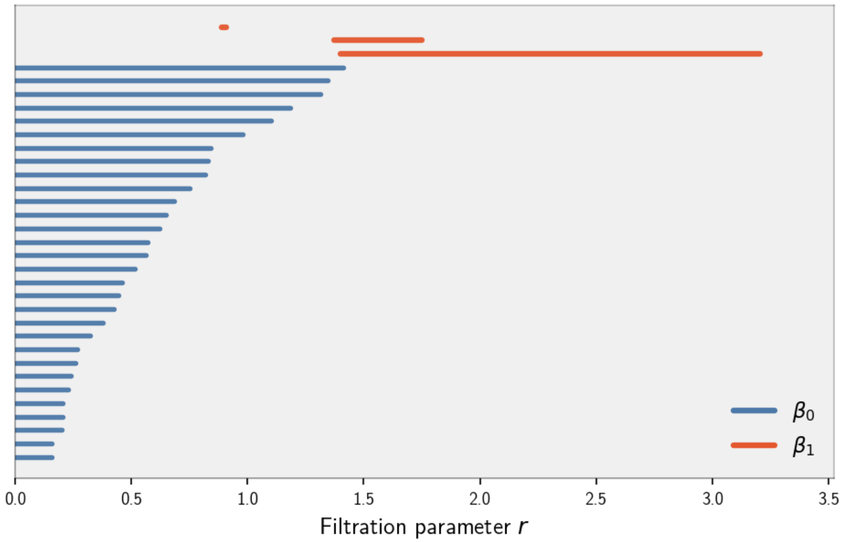
[3]Dashti, A., Asaad, A., Jimenez, M.J., Nanda, V., Paluzo-Hidalgo, E., and Soriano-Trigueros, M. A Survey of Vectorization Methods in Topological Data Analysis. *IEEE Transactions on Pattern Analysis and Machine Intelligen* 45, no. 12 (2023): 14069–80.

Statistical summary of the barcode

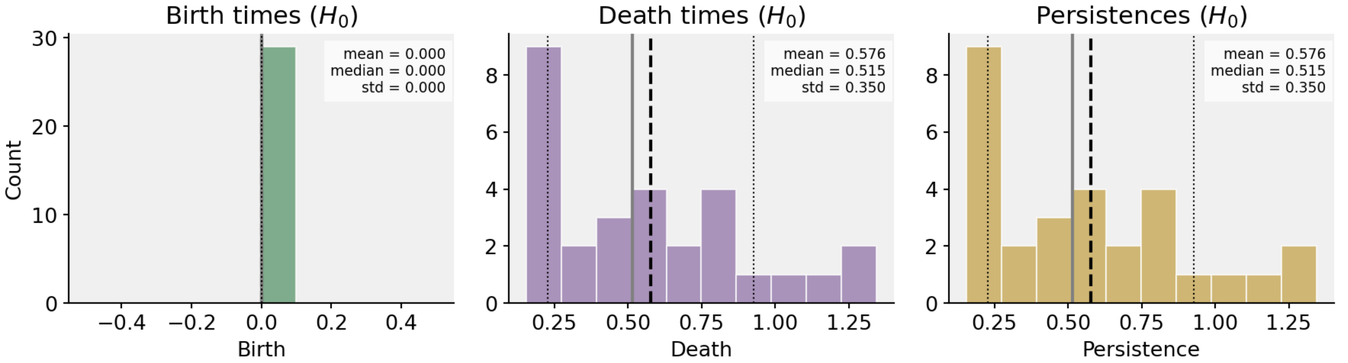
A straightforward vectorization

Define **numerical descriptors** for your barcode:

- Births (vector)
- Deaths (vector)
- Persistences (vector)
- Ratios birth/death (vector)
- Number of bars in the diagram (scalar)
- Total persistence (scalar)
- Persistent entropy[1] (scalar)

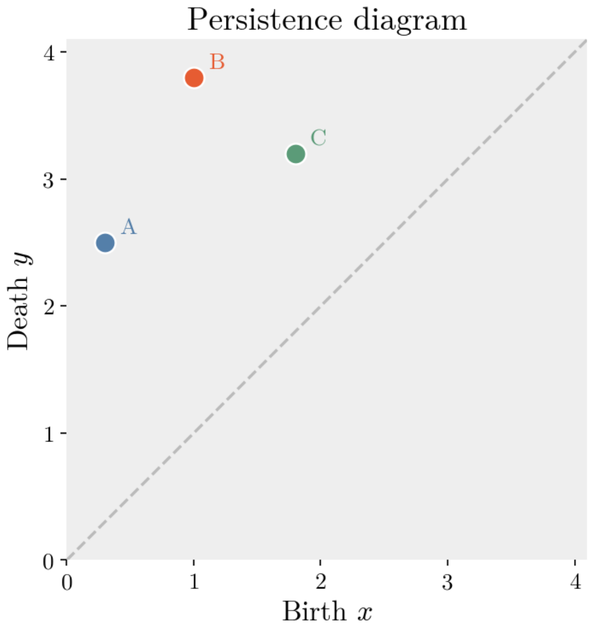


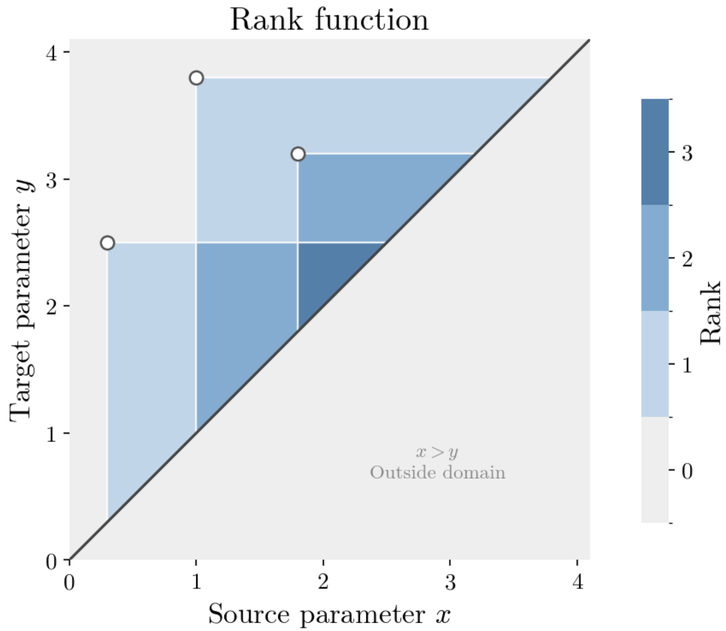
For the vectorial measures, pick some statistics (mean, standard deviation, median, quartiles...) and build a **feature vector** collecting their values for the empirical distributions of these **statistical descriptors of the barcode**



[1]Chintakunta, H., Gentimis, T., Gonzalez-Diaz, R., Jimenez, M. J., and Krim, H. "An Entropy-Based Persistence Barcode." *Pattern Recognition* 48, no. 2 (2015): 391–401.

Rank functions





<u>Def.</u> Rank function

Given a persistence module $M : (\mathbb{R}, \leq) \to \mathrm{Vec}$, its rank function is defined as:

$$\mathrm{rk} : \mathbb{R}^2 \to \mathbb{R}, \quad \mathrm{rk}(x, y) = \begin{cases} \dim \operatorname{im} M(x \leq y) & \text{if } x \leq y, \\ 0 & \text{otherwise}. \end{cases}$$

This is a complete invariant!

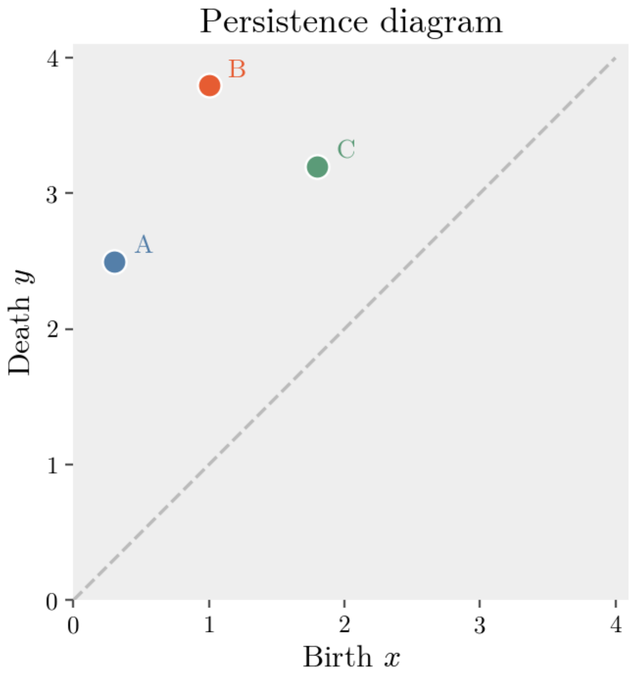
Equivalent to barcodes and diagrams

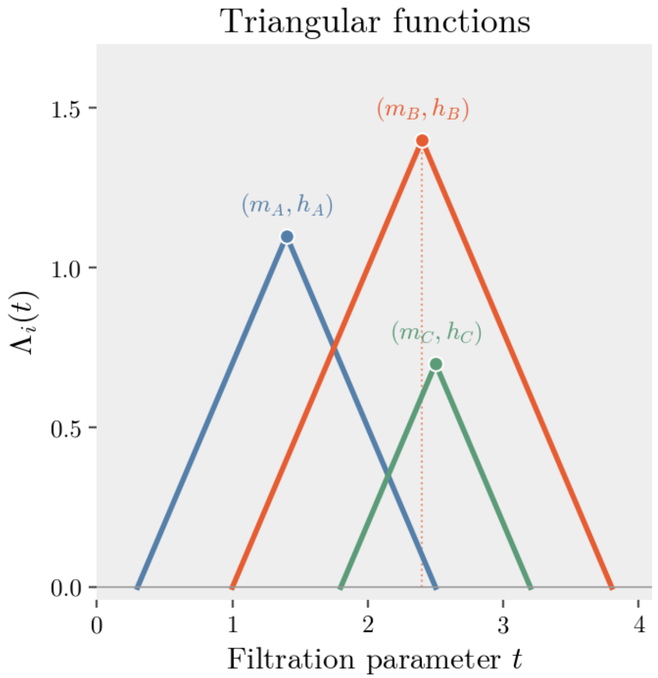
It counts the number of bars at x that stay alive at y

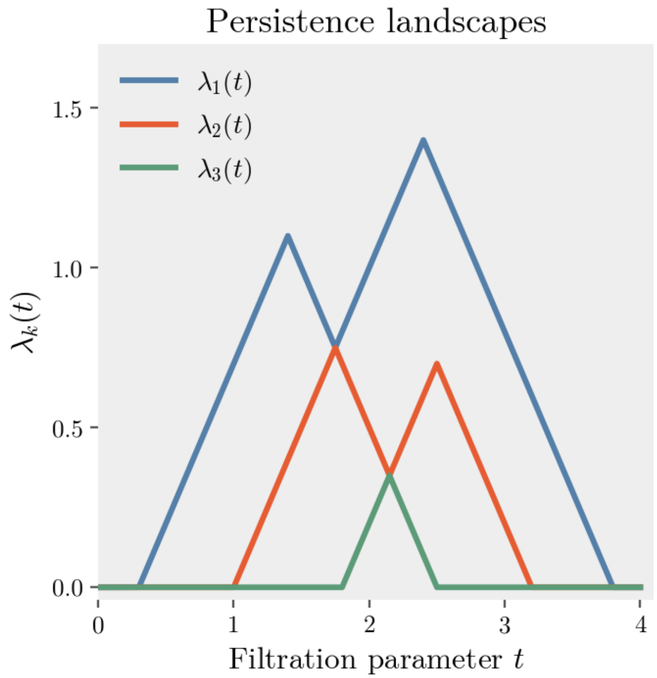
Frosini, Patrizio. A Distance for Similarity Classes of Submanifolds of a Euclidean Space. *Bulletin of the Australian Mathematical Society* 42, no. 3 (1990): 407–15.

Frosini, Patrizio. Discrete Computation of Size Functions. *Journal of Combinatorics, Information & System Sciences* 17, nos. 3–4 (1992): 232–50.

Persistence landscapes







$$(x, y) \longmapsto (m, h) = \left(\tfrac{x+y}{2}, \tfrac{y-x}{2}\right)$$

$$\Lambda_i(t) = \max\{0,\ h_i - |t - m_i|\}$$

$$\lambda_k(t) = \text{kth largest}\{\Lambda_i(t)\}_i$$

<u>Def.</u> Persistence landscape

Given a persistence module $M : (\mathbb{R}, \leq) \to \text{Vec}$, its persistence landscape is defined as:

$$\lambda_k : \mathbb{R} \to \mathbb{R} \cup \{\pm\infty\}, \quad \lambda_k(t) = \sup\{r \geq 0 : \text{rk}\,(t-r, t+r) \geq k\} = k\text{th largest } \{\Lambda_i(t)\}_i$$

Bubenik, Peter. Statistical Topological Data Analysis Using Persistence Landscapes. *Journal of Machine Learning Research* 16, no. 3 (2015): 77–102.

To sum up

Today we introduced the setting of deep learning that we will study in more detail in the two coming days...

And also persistent homology, which is the topological tool that we will use to investigate it

Tomorrow we will see how to incorporate persistence into learning and how to link it to generalization properties of a network

Thank you for your attention!



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